Recap of OO concepts

Objects, classes, methods and more.

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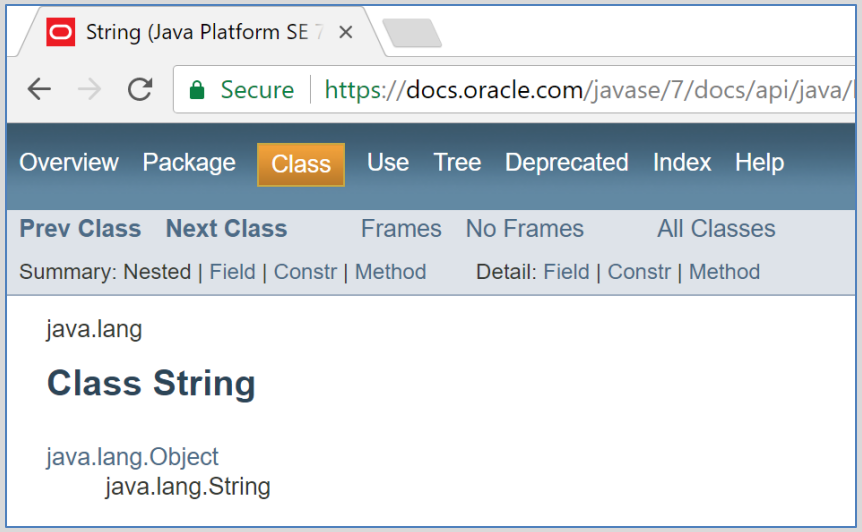
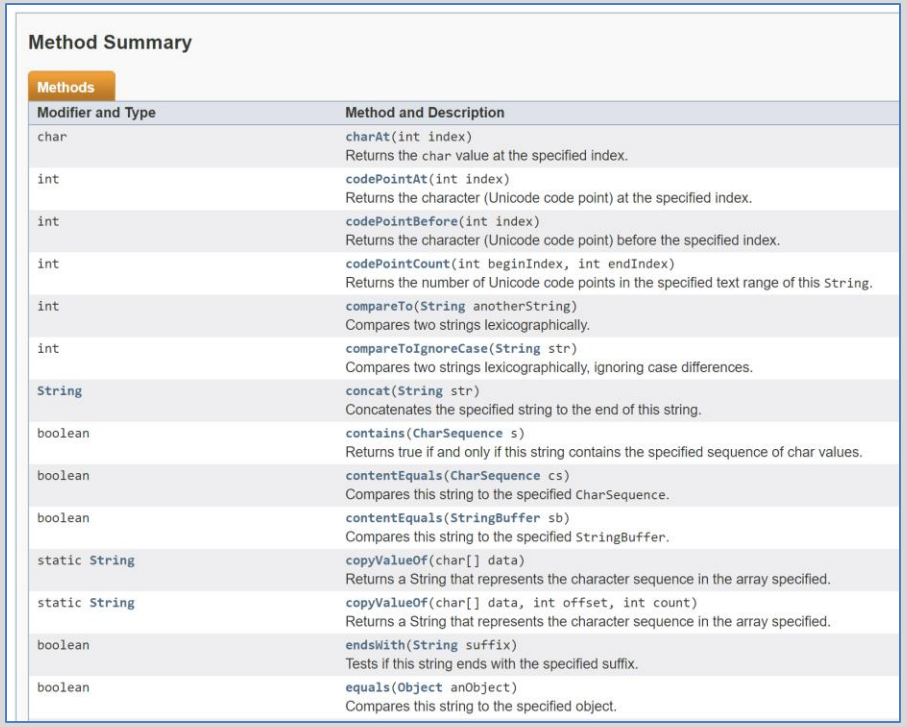
Classes and Objects

- A **class**
  - defines a group of related **methods** (functions) and **fields** (variables / properties).





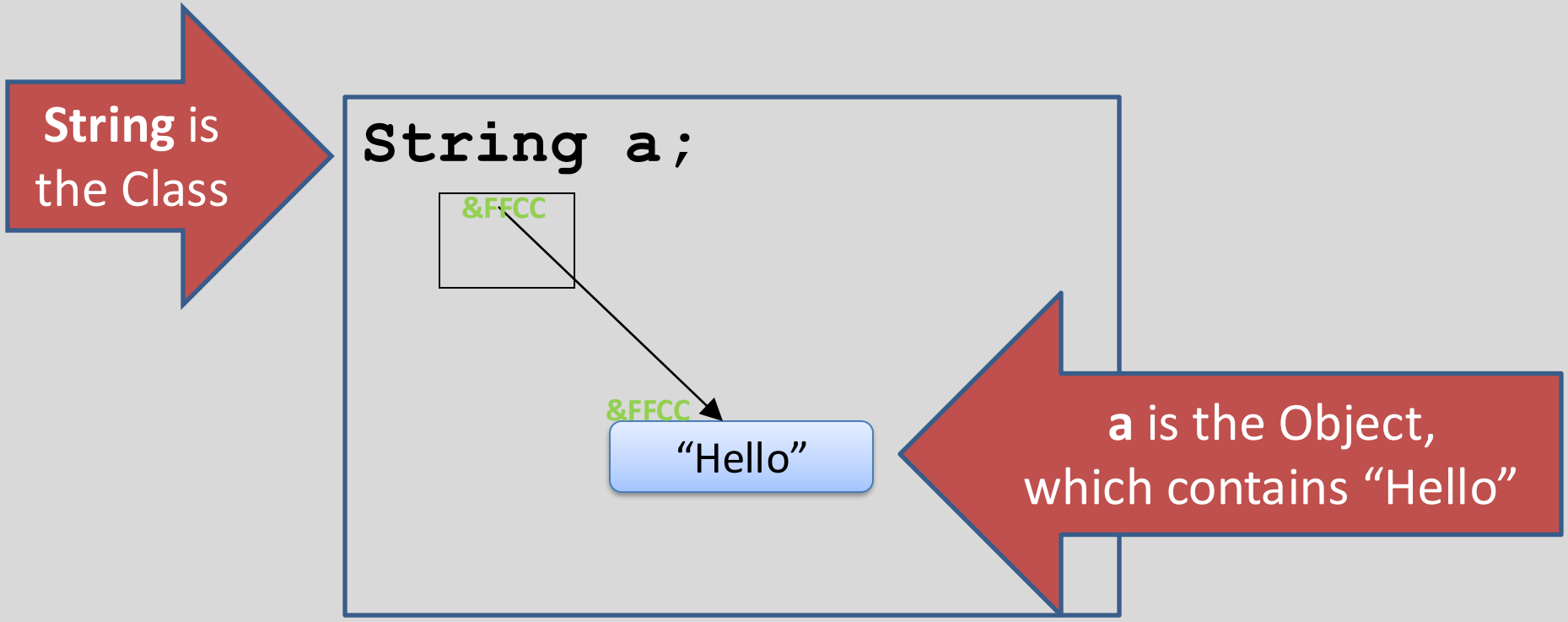
Method Summary

Methods

Modifier and Type	Method and Description
char	charAt(int index) Returns the char value at the specified index.
int	codePointAt(int index) Returns the character (Unicode code point) at the specified index.
int	codePointBefore(int index) Returns the character (Unicode code point) before the specified index.
int	codePointCount(int beginIndex, int endIndex) Returns the number of Unicode code points in the specified text range of this String.
int	compareTo(String anotherString) Compares two strings lexicographically.
int	compareToIgnoreCase(String str) Compares two strings lexicographically, ignoring case differences.
String	concat(String str) Concatenates the specified string to the end of this string.
boolean	contains(CharSequence s) Returns true if and only if this string contains the specified sequence of char values.
boolean	contentEquals(CharSequence cs) Compares this string to the specified CharSequence.
boolean	contentEquals(StringBuffer sb) Compares this string to the specified StringBuffer.
static String	copyValueOf(char[] data) Returns a String that represents the character sequence in the array specified.
static String	copyValueOf(char[] data, int offset, int count) Returns a String that represents the character sequence in the array specified.
boolean	endsWith(String suffix) Tests if this string ends with the specified suffix.
boolean	equals(Object anObject) Compares this string to the specified object.

Classes and **Objects**

- An **object**
  - is a single instance of a class
  - i.e. an object is created (instantiated) from a class.



Classes and Objects – **Many Objects**

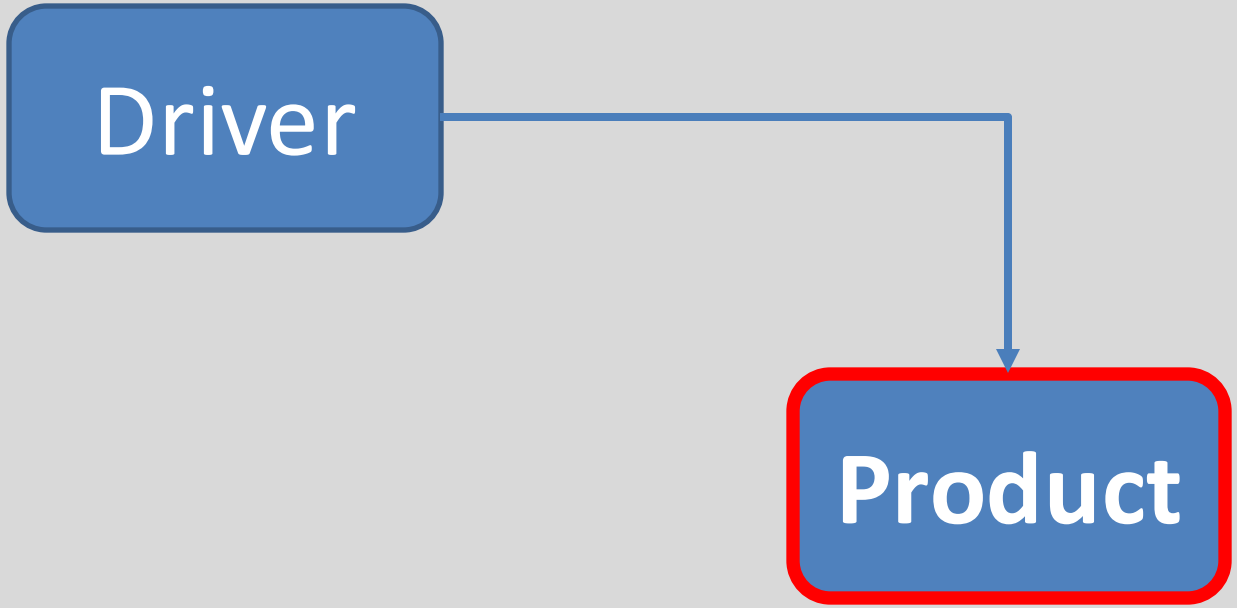
- Many **objects** can be constructed from a single **class** definition.
- Each **object** must have a unique name within the program.

Ver 1.0

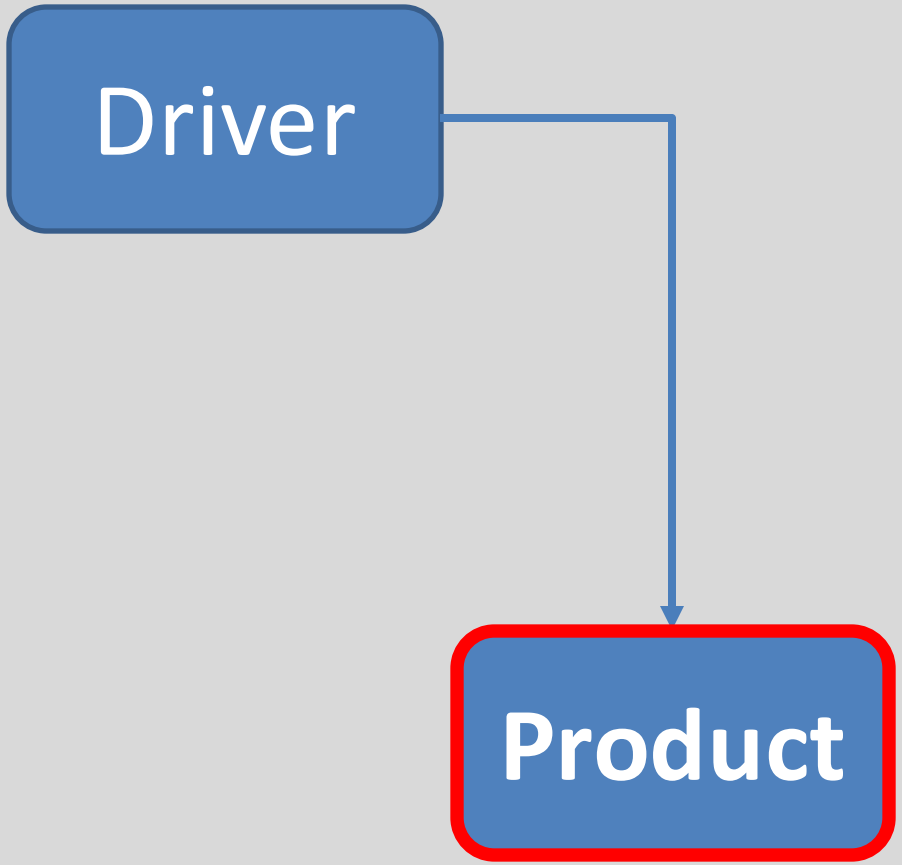
SHOP

Shop V1.0 - **Product**

- We will recap object oriented concepts through the study of a new class called **Product**.

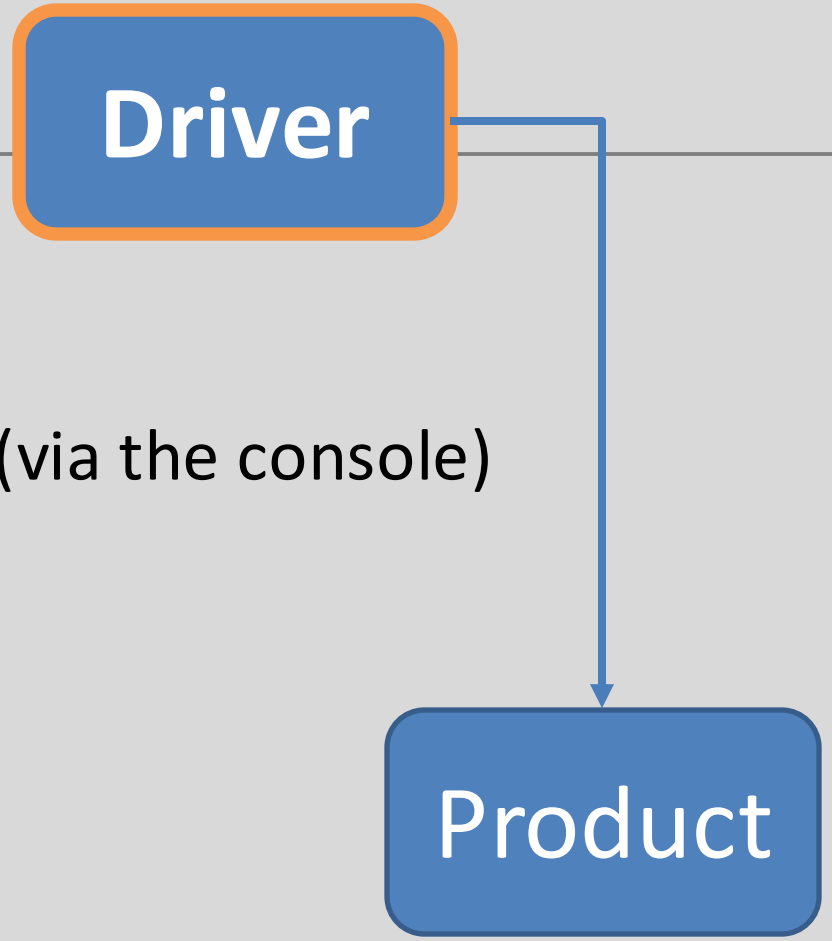


Shop V1.0 - **Product**



- The **Product** class stores **details** about a product
  - name
  - code
  - unit cost
  - in the current product line or not?

Shop V1.0 - **Driver**



- The Driver class
  - has the **main()** method.
  - **reads** the product details from the user (via the console)
  - **creates** a new Product object.
  - **prints** the product object (to the console)

- Driver is covered in the next lecture.

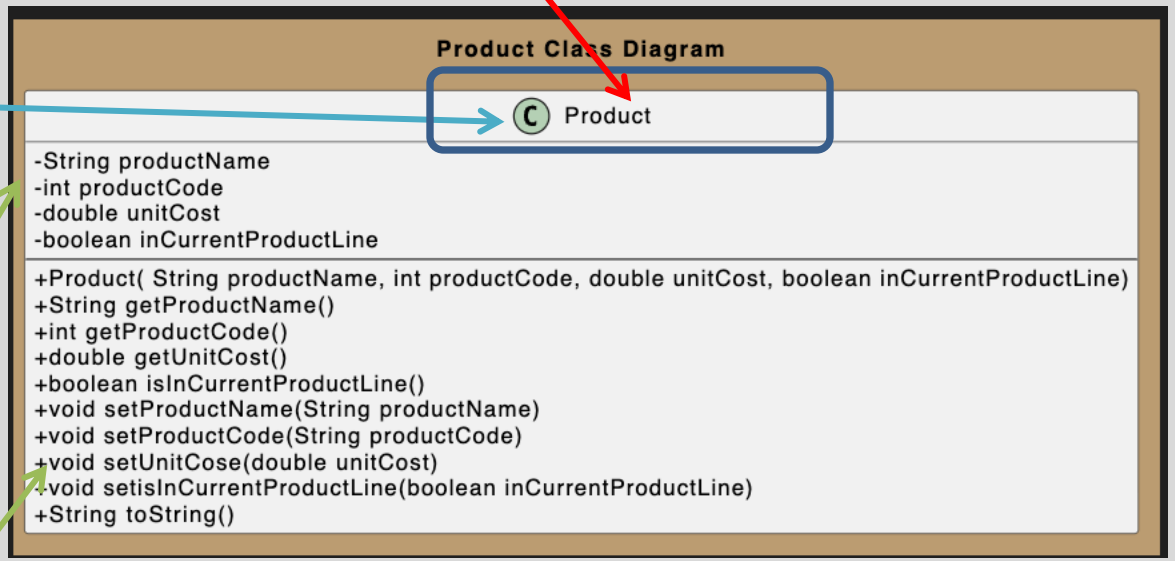
A **Product** Class...

Object Type/ **Class** Name
i.e. Product

The **C** icon means it is a **Class**.

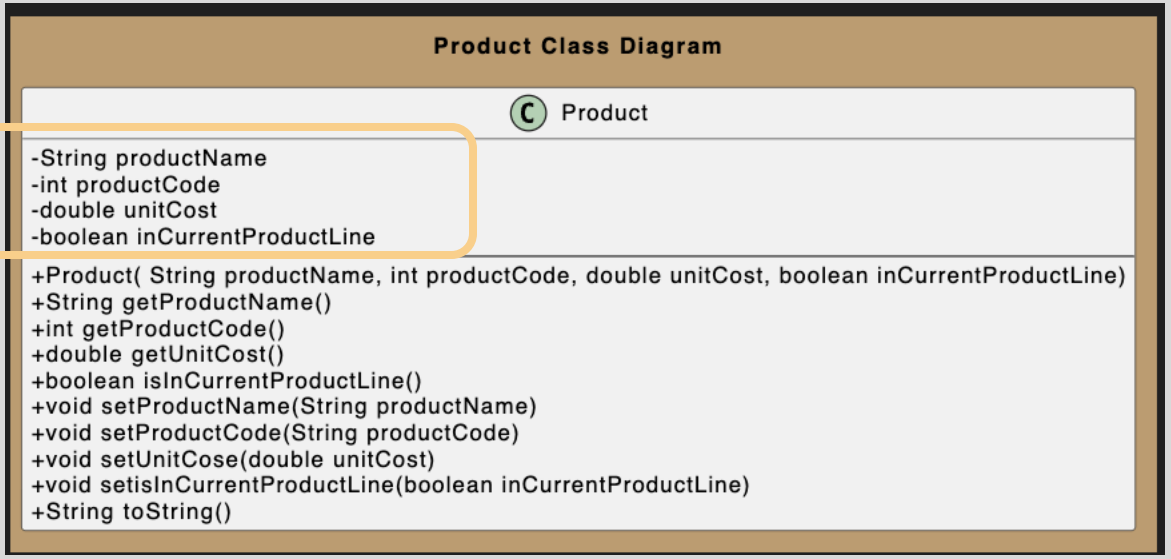
The '-' means it is **private**.

The '+' means it is **public**.



A Product Class…**fields**

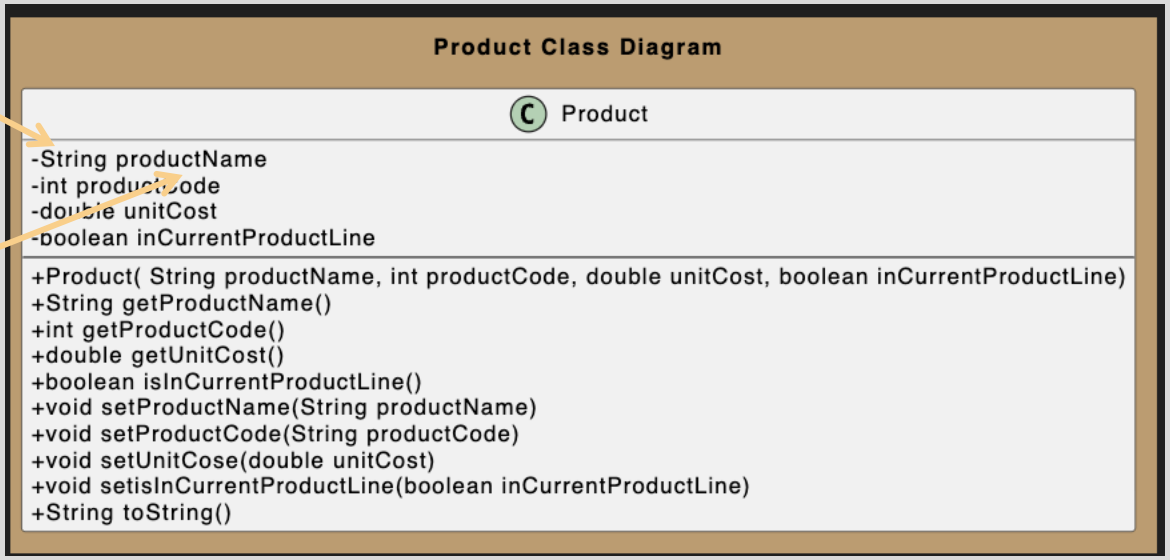
Fields
i.e. the **attributes / properties** of the class



A Product Class…**fields**

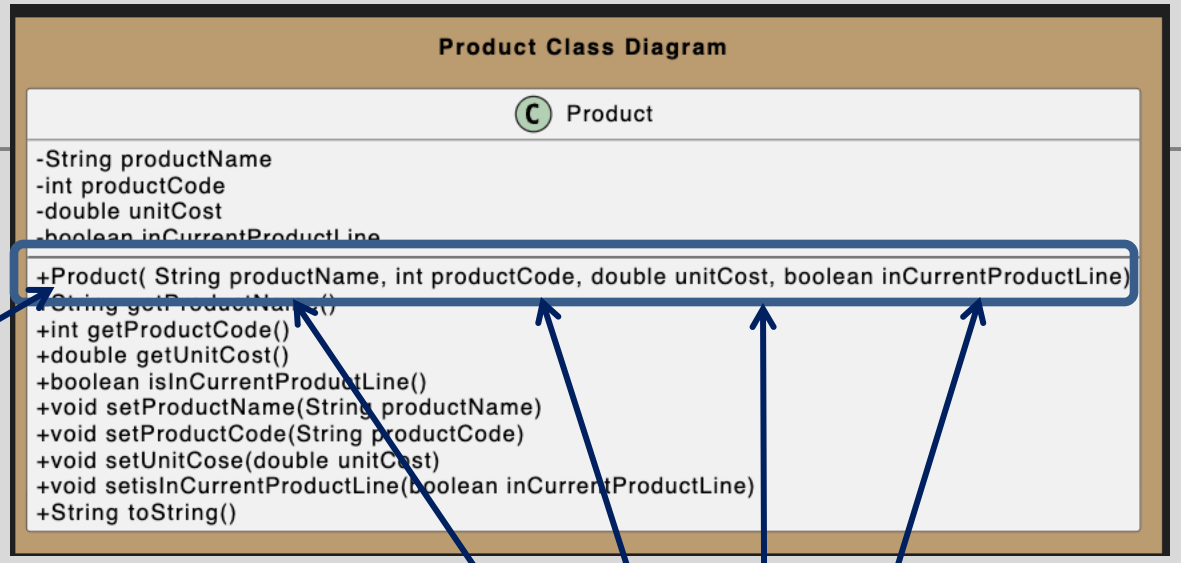
field **type**

field **name**



A **Product** Class…
Constructor

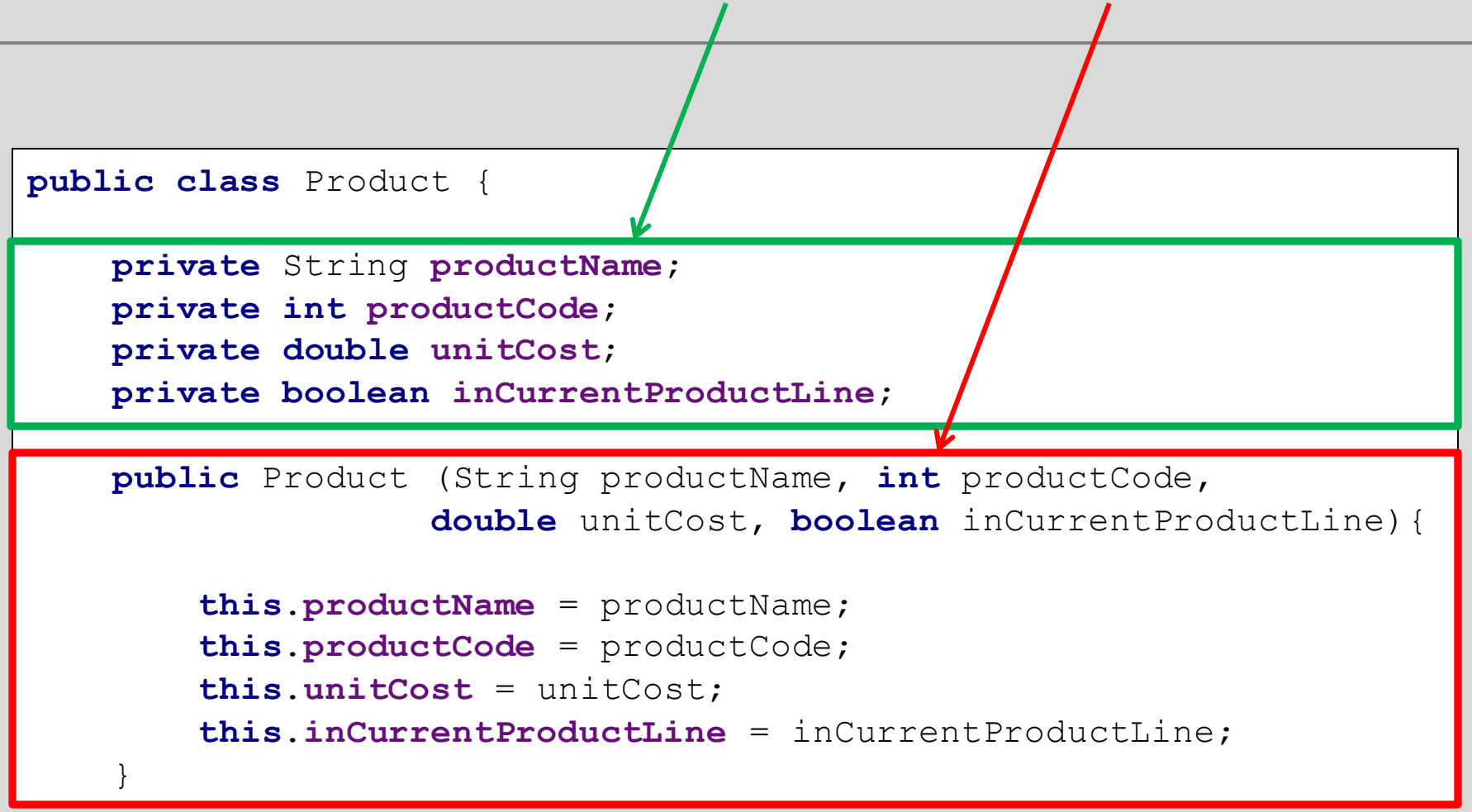
Constructor
i.e. for building objects.



Constructors have same name as the class

Four **parameters**;
one for each field.

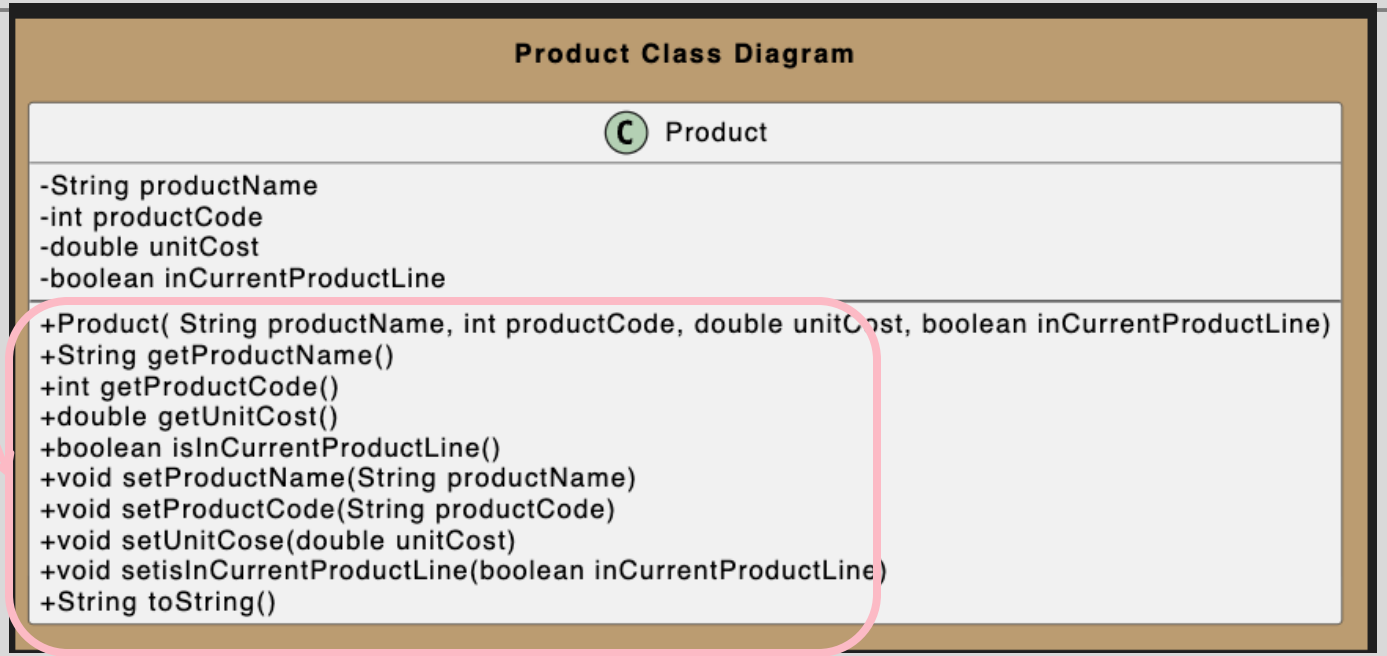
A **Product** Class… **fields** and **constructor**



```
public class Product {

    private String productName;
    private int productCode;
    private double unitCost;
    private boolean inCurrentProductLine;

    public Product (String productName, int productCode,
                    double unitCost, boolean inCurrentProductLine){

        this.productName = productName;
        this.productCode = productCode;
        this.unitCost = unitCost;
        this.inCurrentProductLine = inCurrentProductLine;
    }
```

A Product Class… **methods**

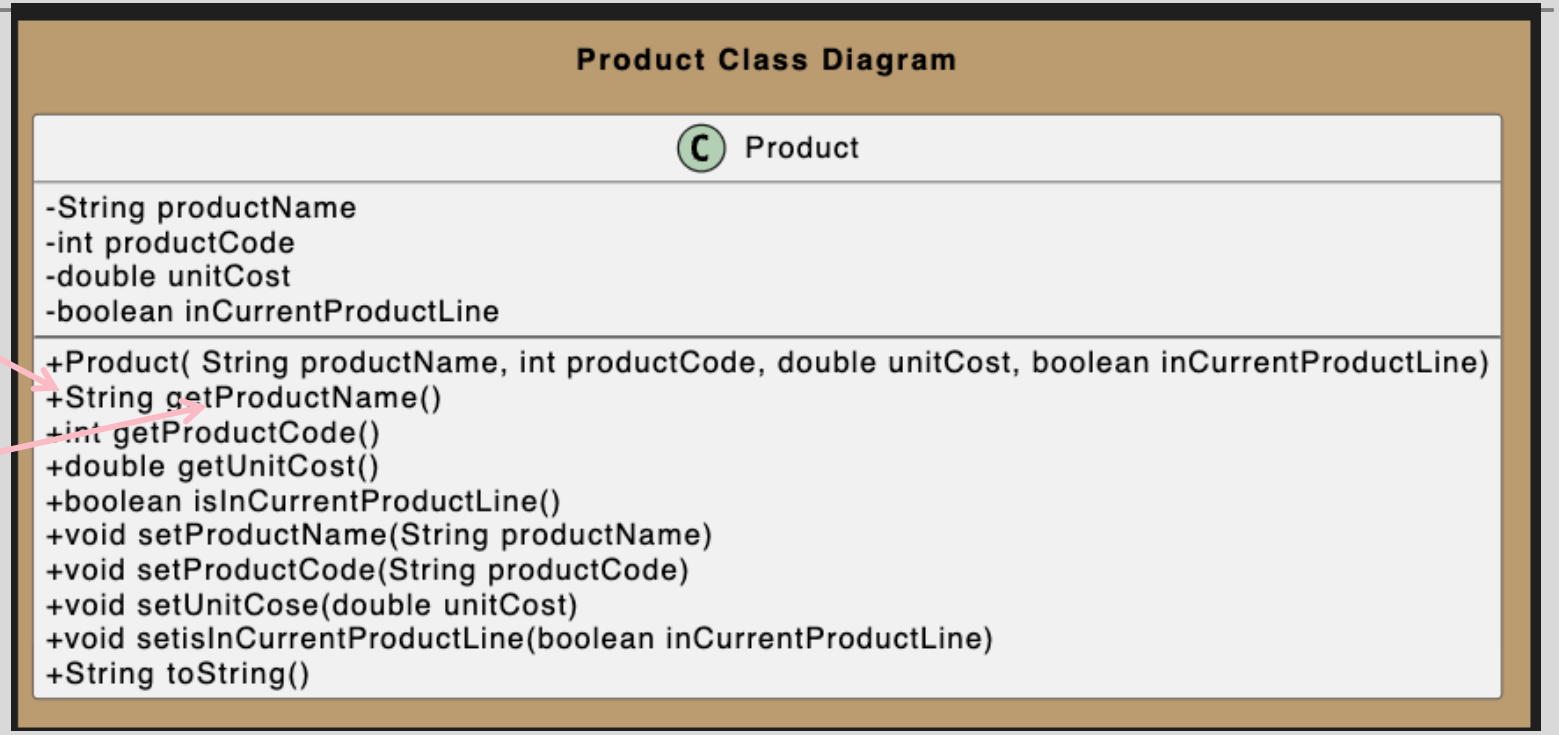
Methods
i.e. the **behaviours** of the class



A Product Class… **methods**

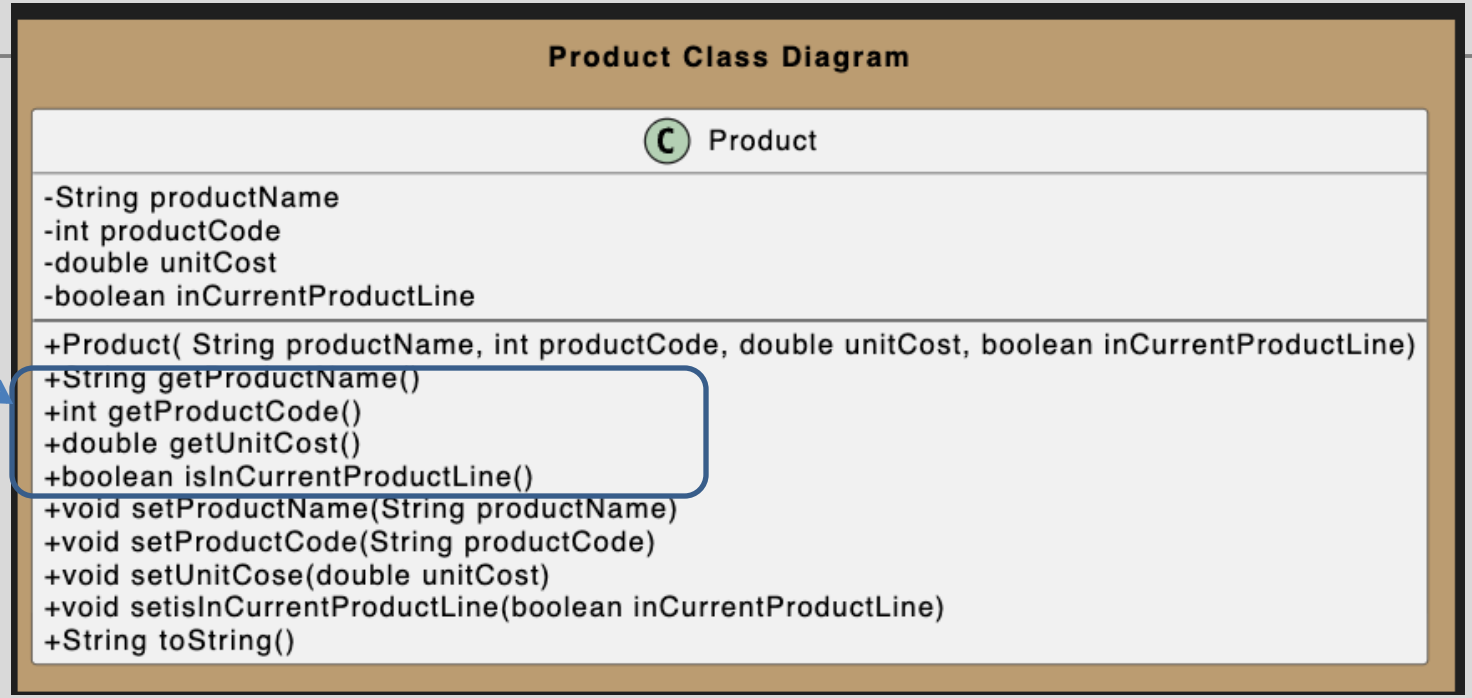
Return type

Method **name**



A Product Class… **get**ters

getters

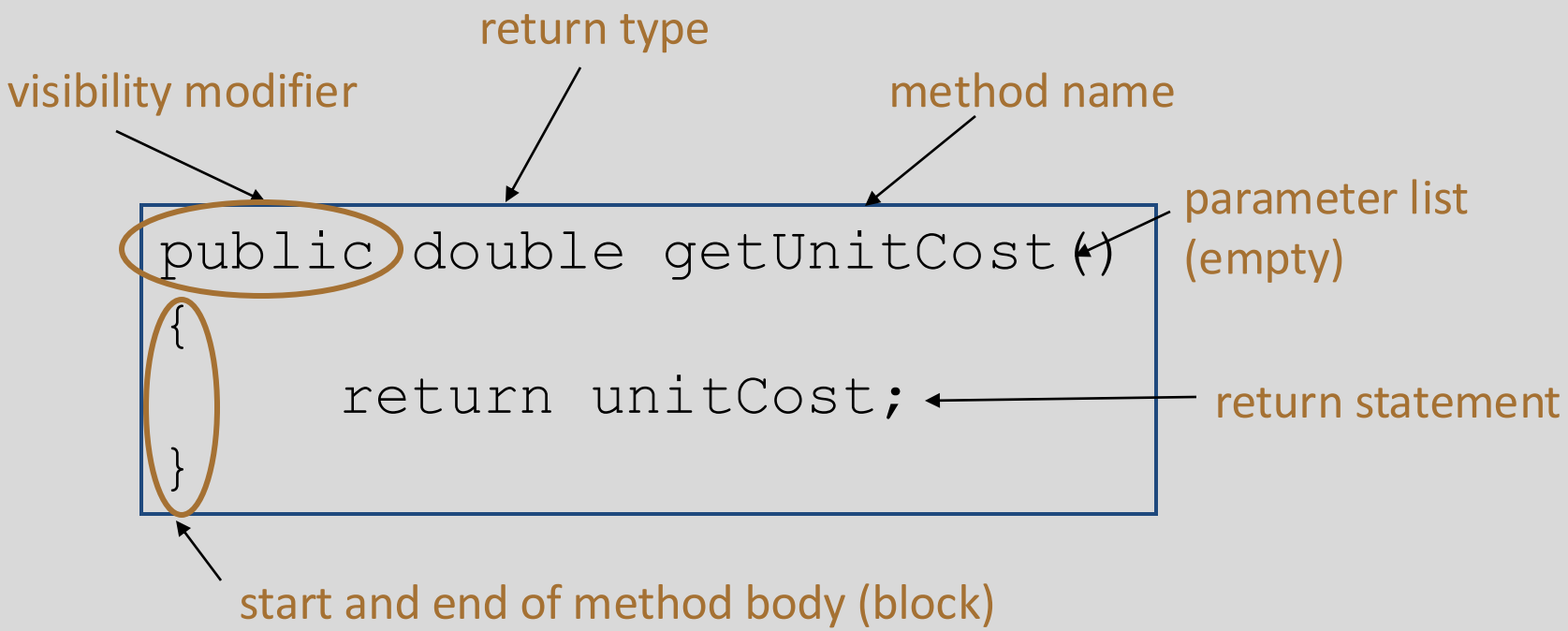


Getters (Accessor Methods)

- **Accessor** methods
  - return information about the **state** of an object
    - i.e. **the values stored in the fields**.

- A 'getter' method
  - is a specific type of **accessor** method and typically:
    - **contains a return statement**
      (as the last executable statement in the method).
    - defines a **return type**.
    - **does NOT change the object state**.

Getters

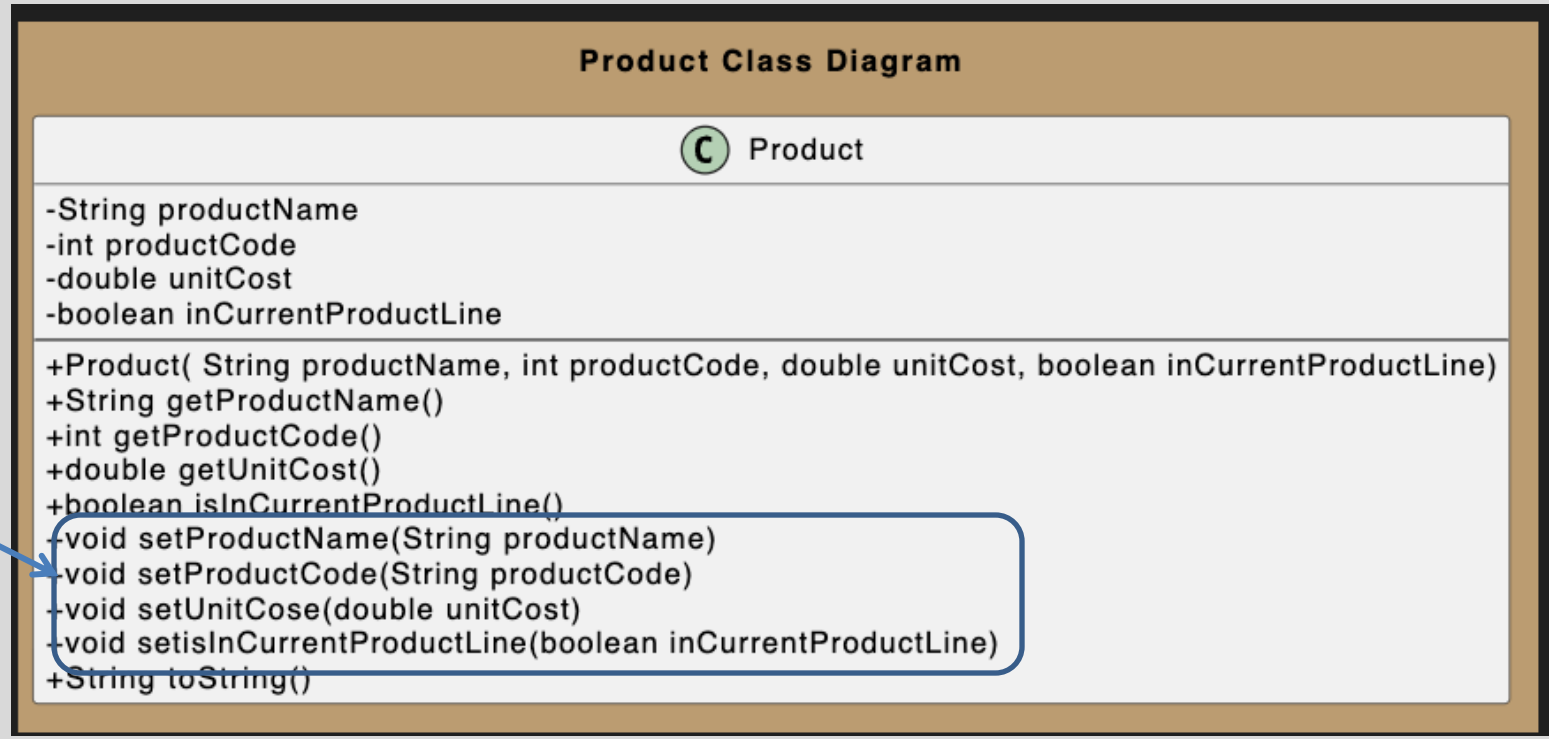

return type
visibility modifier
method name
parameter list
(empty)
public double getUnitCost()
{
return unitCost;
}
return statement
start and end of method body (block)

A Product Class…**get**ters

```
public String getProductName(){
    return productName;
}

public double getUnitCost(){
    return unitCost;
}

public int getProductCode() {
    return productCode;
}

public boolean isInCurrentProductLine() {
    return inCurrentProductLine;
}
```

A Product Class…**set**ters

setters

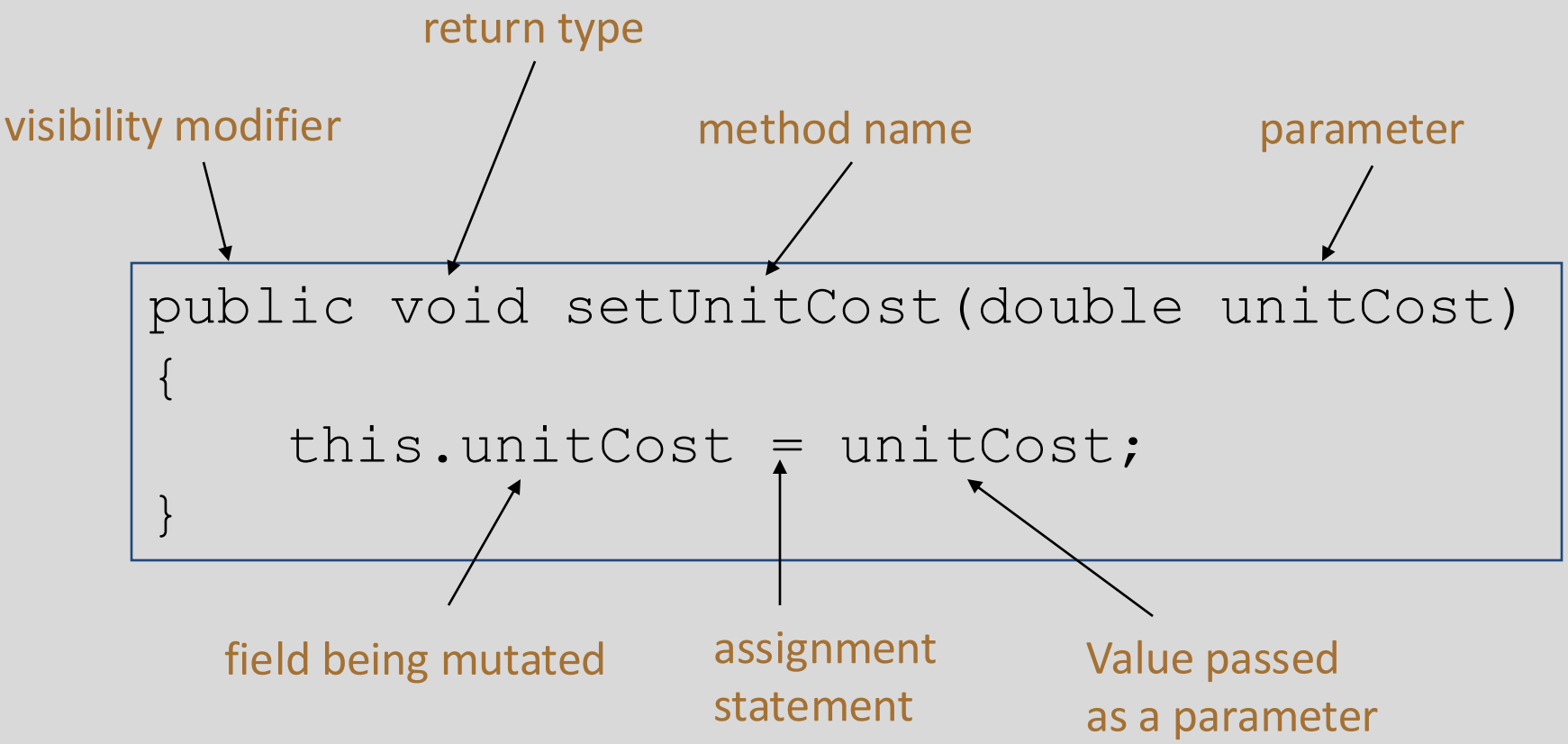


Setters (Mutator methods)

- **Mutator** methods
  - change (i.e. mutate!) an object's state.

- A 'setter' method
  - is a specific type of **mutator** method and typically:
    - contains an **assignment statement**
    - takes in a **parameter**
    - **changes the object state**.

Setters


return type
visibility modifier
method name
parameter
public void setUnitCost(double unitCost)
{
this.unitCost = unitCost;
}
field being mutated
assignment statement
Value passed as a parameter

A Product Class…setters

```
public void setProductCode(int productCode) {
    this.productCode = productCode;
}

public void setProductName(String productName) {
    this.productName = productName;
}

public void setUnitCost(double unitCost) {
    this.unitCost = unitCost;
}

public void setInCurrentProductLine(boolean inCurrentProductLine) {
    this.inCurrentProductLine = inCurrentProductLine;
}
```

Getters/Setters

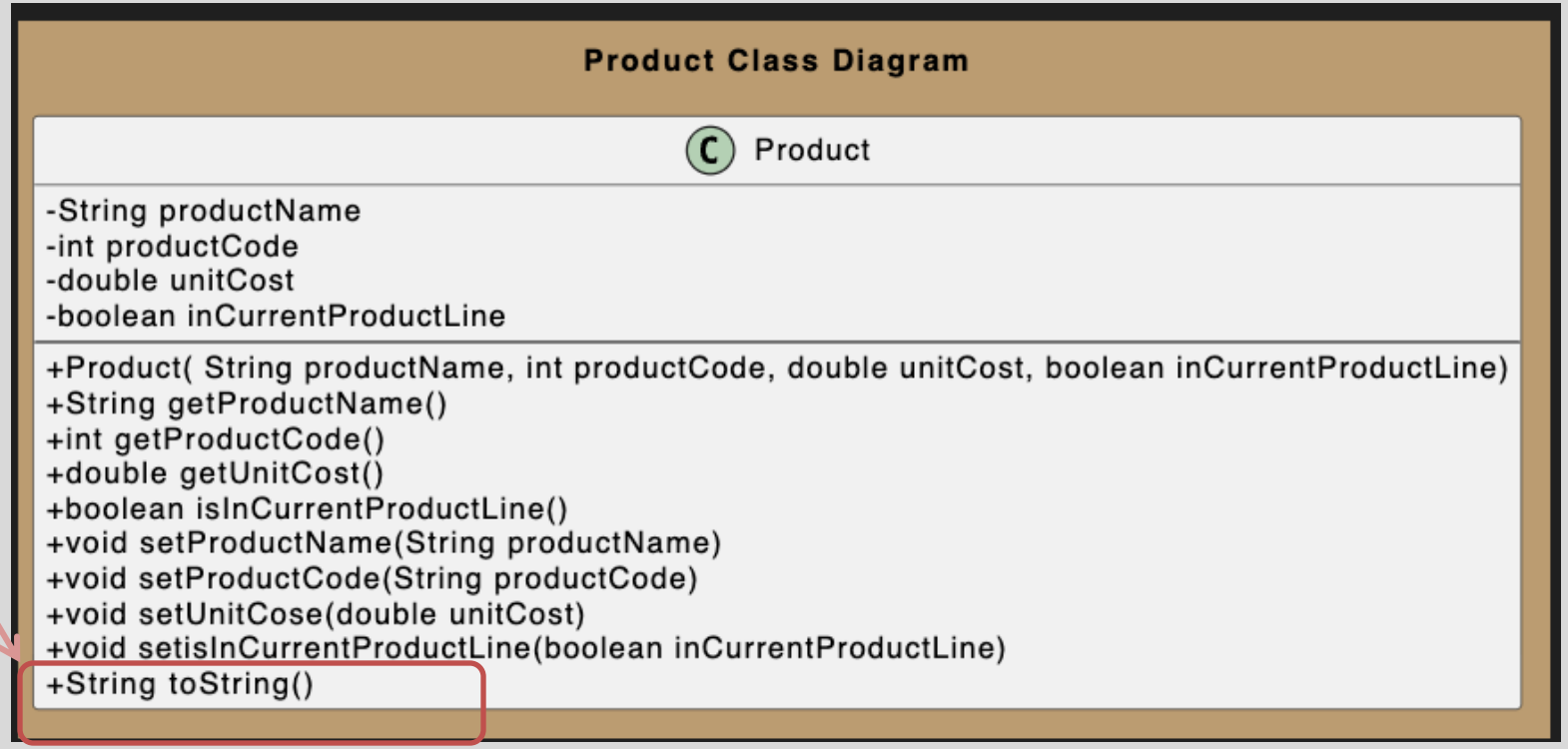
For each instance field in a class, you are normally asked to write:

- A **get**ter
  - Return statement
- A **set**ter
  - Assignment statement

A Product Class…**toString**

toString():

Builds and returns a String containing a user-friendly representation of the object state.



A Product Class…

```
public String toString()
{
     return "Product description: " + productName
          + ", product code: " + productCode
          + ", unit cost: " + unitCost
          + ", currently in product line: " + inCurrentProductLine;
}
```

Sample Console Output if we printed a Product Object:

```
Product description: 55 Inch TV, product code: 23432, unit cost: 1399.99, currently in product line: true
```

toString()

- This is a useful method and you will write a **toString**() method for most of your classes.
- **When you print an object, Java automatically calls the toString() method**
  e.g.

```
Product product = new Product();

//both of these lines of code do the same thing
System.out.println(product);
System.out.println(product.toString());
```

Encapsulation in Java – **steps 1-3**

Encapsulation Step	Approach in Java

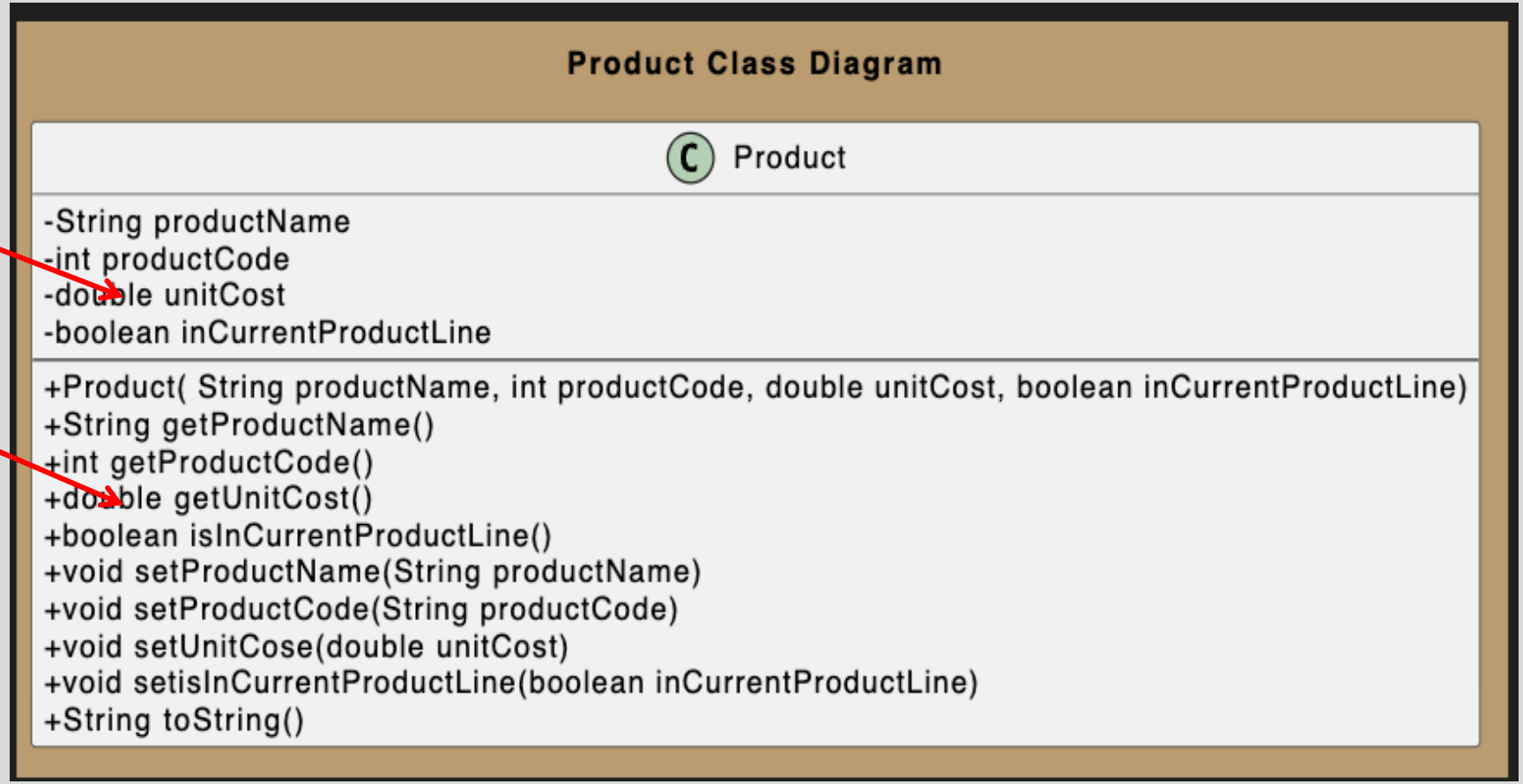
1. **Wrap** the data (fields) and code acting on the data (methods) together as single unit.	`public class ClassName`
	`{`
	`  Fields`
	`  Constructors`
	`  Methods`
	`}`

2. **Hide** the fields from other classes.	**Declare the fields of a class as private.**
3. **Access** the fields only through the methods of their current class.	**Provide public setter and getter methods** to modify and view the fields values.

http://www.tutorialspoint.com/java/java_encapsulation.htm

A Product Class… **An Encapsulated Class**

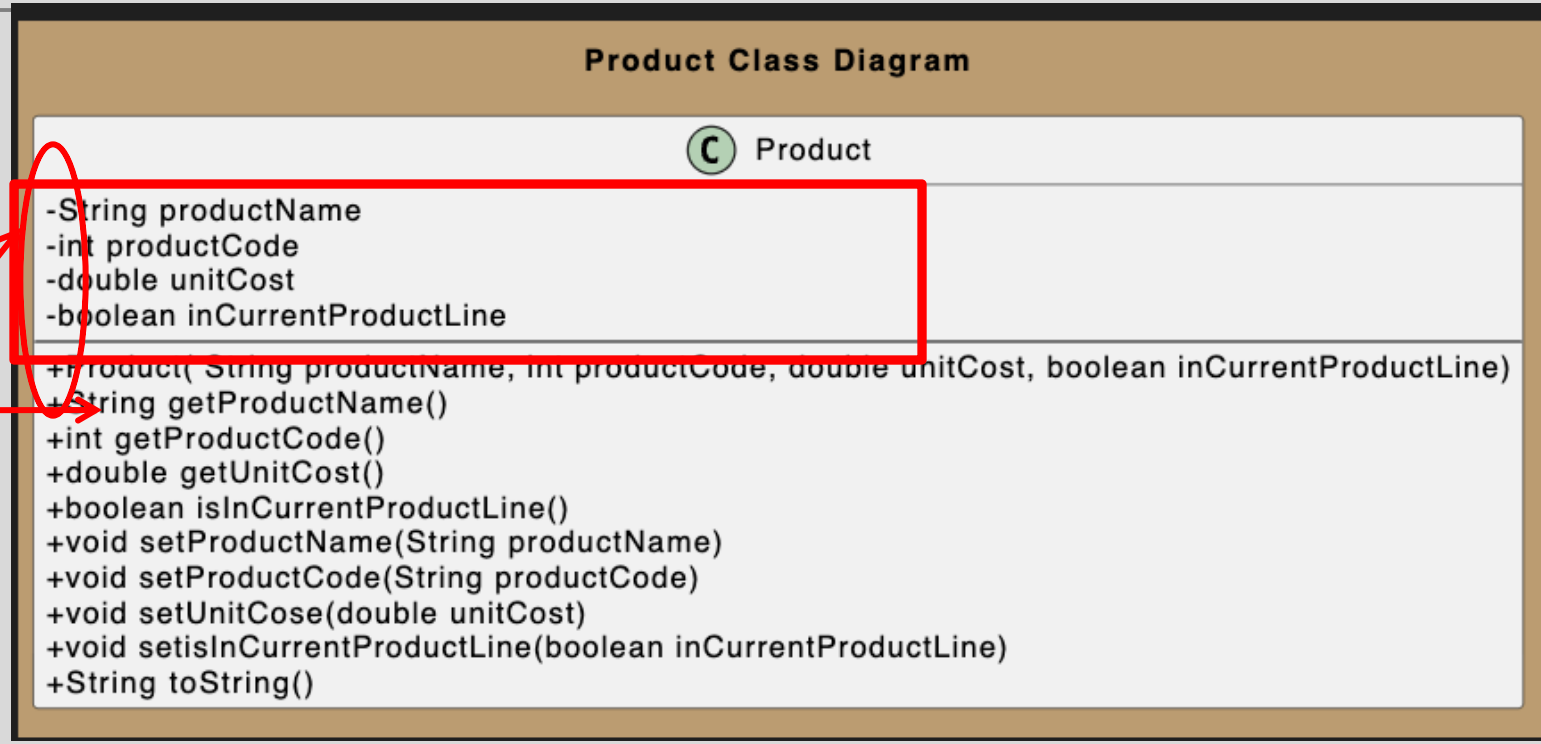
1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.



A Product Class… **An Encapsulated Class**

1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.

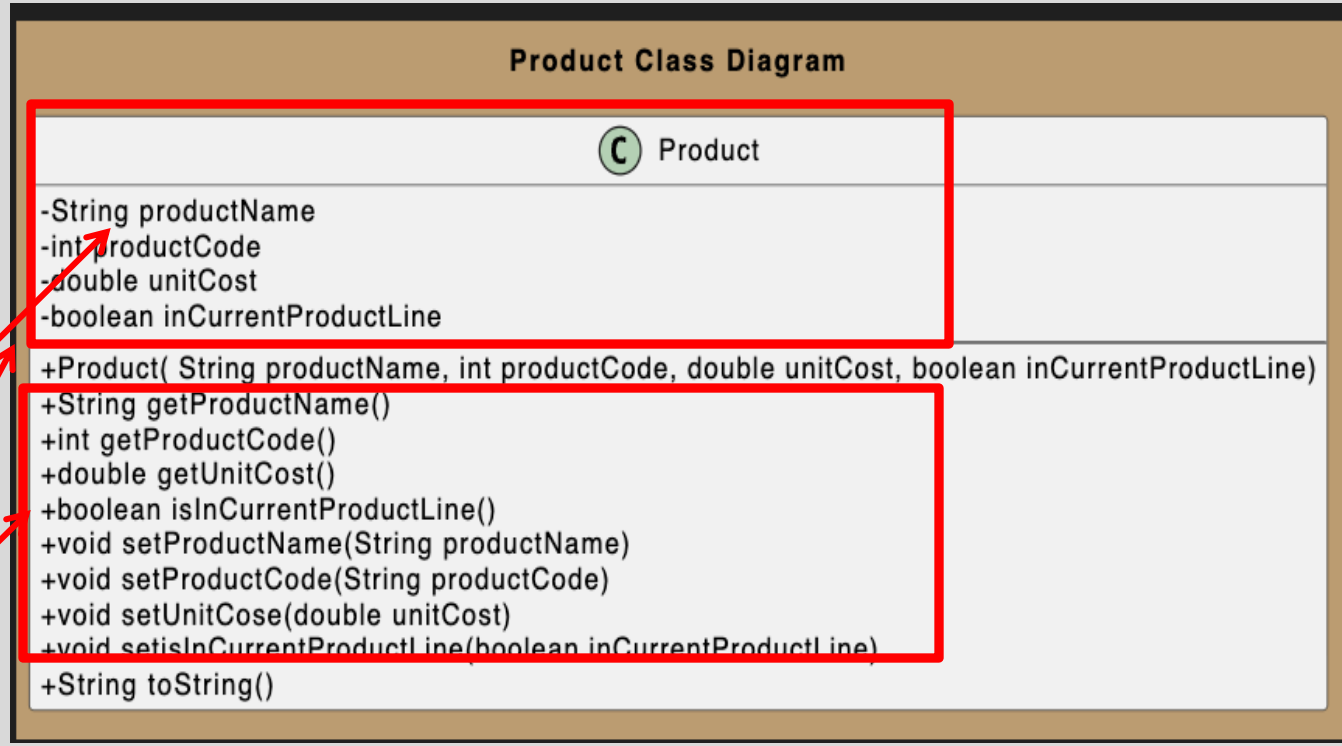


A Product Class… **An Encapsulated Class**

1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.

3. **Access** the fields only through the methods of Product (e.g. **get**ter and **set**ter methods).



Using the Product Class

1

```
private Product product;
```

Declaring an object **product**, of type **Product**.

product

null

Using the Product Class

1

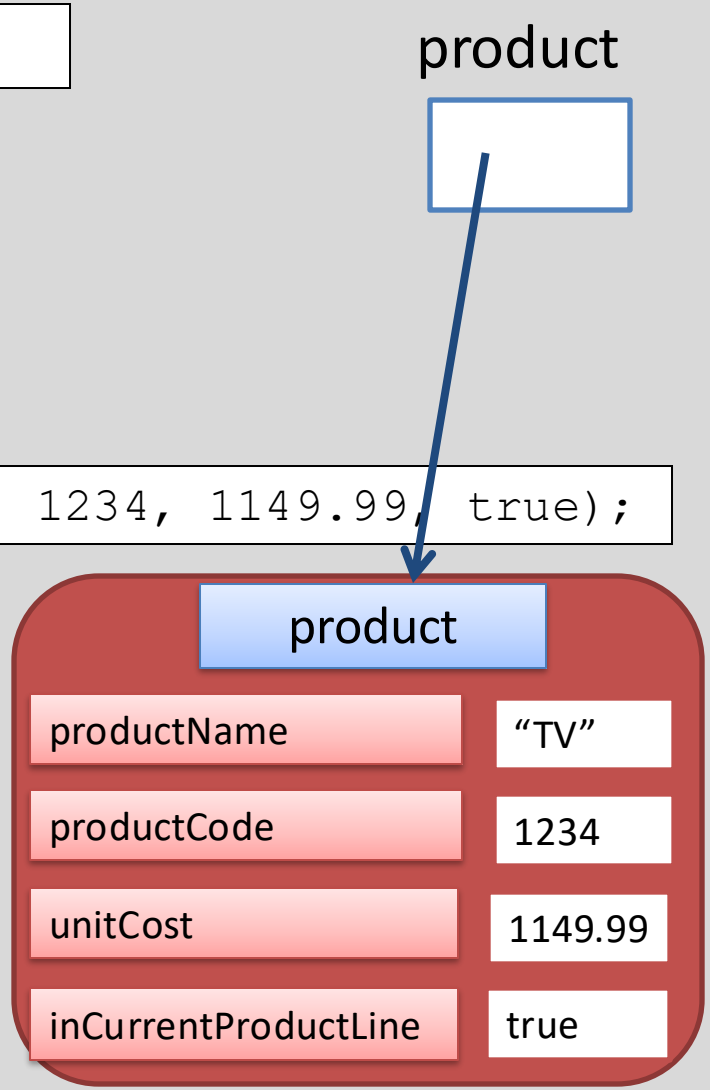
```
private Product product;
```

Declaring an object **product**, of type **Product**.

2

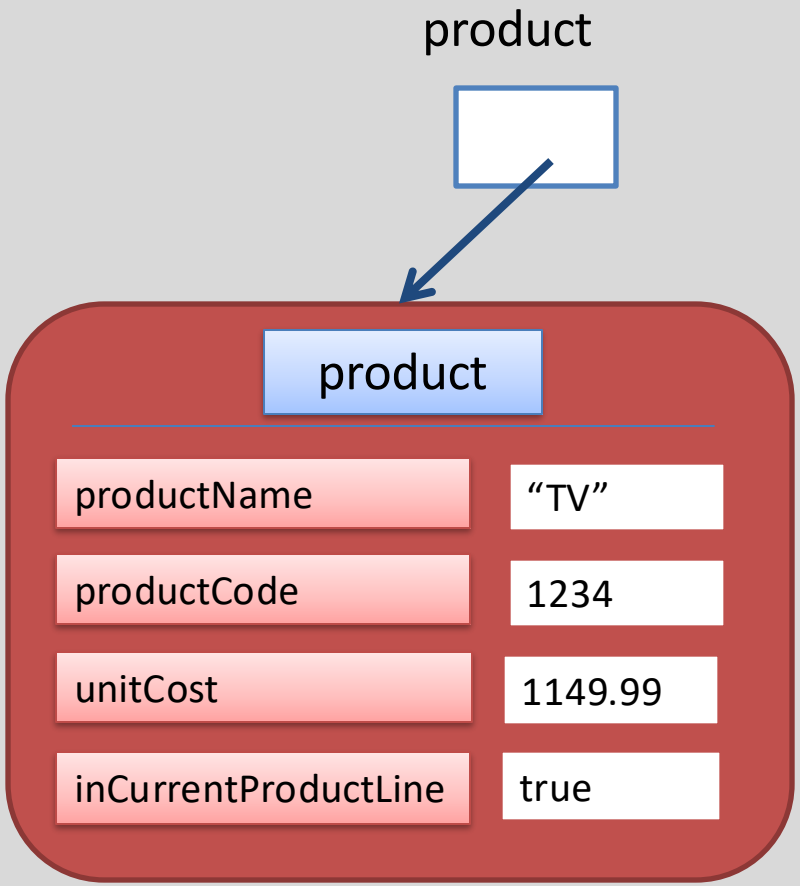
```
product = new Product("TV", 1234, 1149.99, true);
```

Calls the **Product** *constructor* to build the **product** object in memory.



Multiple Product Objects

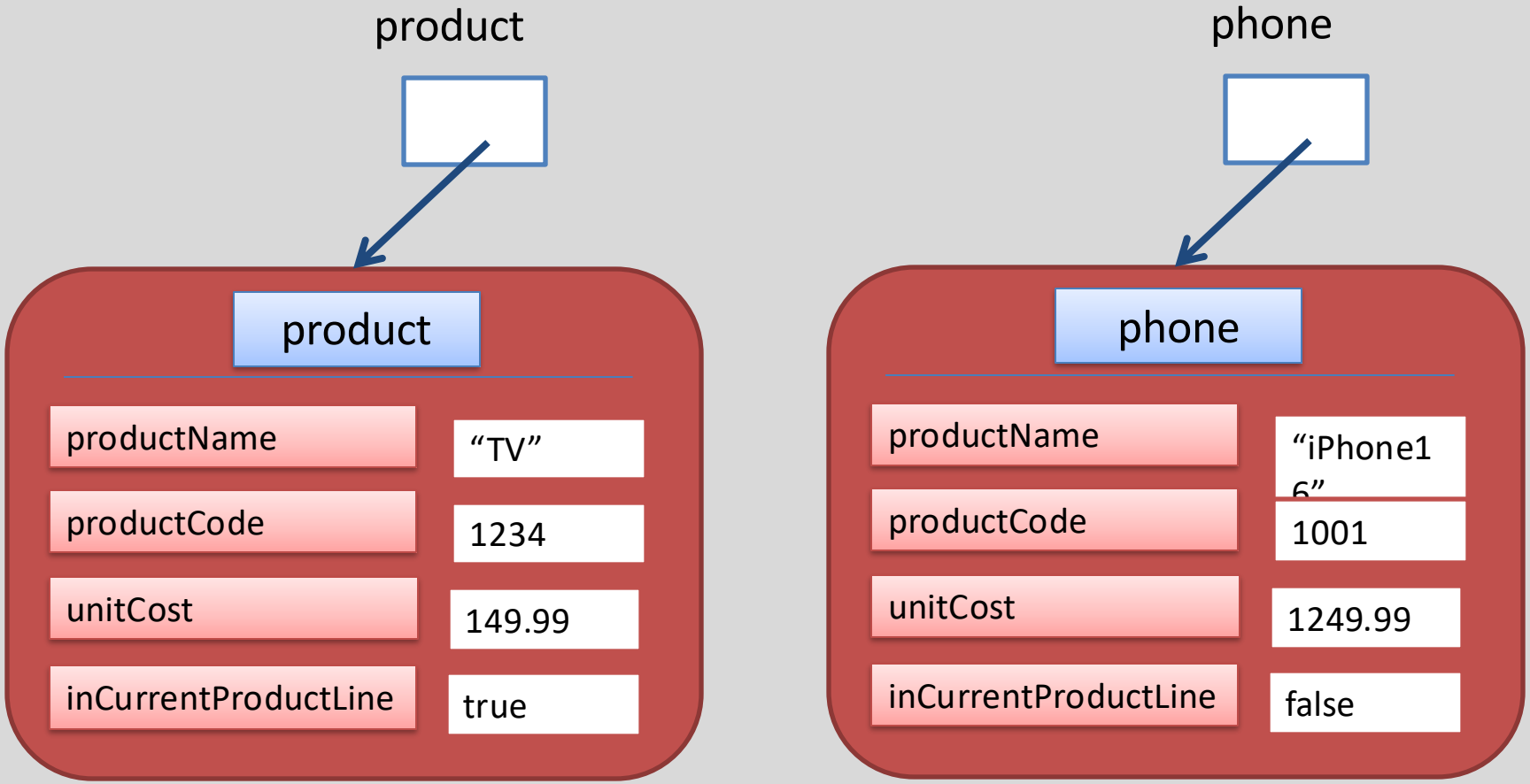
```
private Product product = new Product("TV", 1234, 1149.99, true);
```



Multiple Product Objects

```
private Product product = new Product("TV", 1234, 149.99, true);
```

```
private Product phone = new Product("iPhone16", 1001, 1249.99, false);
```



Questions?